← Back **Check your understanding: End of module 1**
Graded Assignment • 20 min

Due Jul 1, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. In geometry, what is the formula for the area of a circle? 1 / 1 point

- ( ) $A = 2\pi r$
- ( ) $A = \pi d$
- (•) $A = \pi r^2$
- ( ) $A = \pi r^3$

Nice work

Great job! Remember that r is the radius of the circle and π is a constant representing the ratio of the circumference to the diameter.

2. Which of the following are true about the unit circle? 1 / 1 point

- [x] Sine values correspond to the y-coordinate.

  Nice work

  Well done! The sine function is tied to the vertical coordinate on the unit circle.

- [x] The radius is 1.

  Nice work

  Good! The unit circle's definition is based on this radius.

- [x] The circumference is

  $$2\pi$$

  .

  Nice work

  Good! This is a fundamental identity for the unit circle.

- [x] Angles are measured in radians.

  Nice work

  Correct! Radians and degrees are used, but radians are conventional here.

- [x] Cosine values correspond to the x-coordinate.

  Nice work

  Correct! The cosine function relates to the x-axis coordinates.

- [ ] The center is at (1,0).

3. What is the derivative of $f(x) = x^3$ with respect to x? 1 / 1 point

- ( ) $f'(x) = 2x^3$
- (•) $f'(x) = 3x^2$
- ( ) $f'(x) = x^2$
- ( ) $f'(x) = 3x^3$

Nice work

Excellent! You correctly applied the power rule for derivatives.

4. Select the properties that correctly describe a circle in the two-dimensional plane. 1 / 1 point

- [ ] A circle has infinite sides.
- [ ] The circumference is twice the diameter.
- [x] The area is determined by

  $$\pi r^2$$

  .

  Nice work

  Well done! The area formula relates directly to the radius.

- [x] All radii are equal in length.

  Nice work

  Correct! This is a fundamental property of circles.

- [x] The diameter is the longest chord.

  Nice work

  Correct! The diameter is a special chord that passes through the center.

5. Which of the following are applications of calculus in computing? 1 / 1 point

- [x] Optimization problems.

  Nice work

  Excellent choice! Calculus is often used to solve optimization problems in computing.

- [x] Performance analysis and modeling.

  Nice work

  Well done! Performance analysis often uses calculus for modeling and evaluation.

- [ ] Database management systems.
- [x] Image processing algorithms.

  Nice work

  Image processing can utilize calculus to analyze and manipulate visual data effectively.

- [ ] Cryptography.